List of experiments

Lab No.	Objectives	Pre-requisites	Ref.
L1	• To get familiar with basic supported data structures, including lists, tuples, dictionaries, etc. • Familiar with basic Python packages numpy, pandas, etc.	Pandas	
L2	• Data preprocessing • Basic data visualization using Matplotlib package.	Matplotlib	
L3	• Line graph • Scatter plot	Matplotlib	
L4	• Pie Chart • Violin Plot • Heat map	Matplotlib	
L5	• Subplots • Coordinate systems	Matplotlib	
L6	• Analyse population data through visualization	Matplotlib	
L7	• Figure, figure size, subplots, ticks, grid, markers, legends, axes, scale, save quality image.	Matplotlib	
L8	• Use of color, text, and annotation.	Matplotlib	
L9	• Use of 3-D plotting	Matplotlib	
L10	Complex data visualization • Time series data • Images data • Geospatial Data	Matplotlib	

Resources:

1. https://github.com/ozlerhakan/datacamp
2. https://machinelearningmastery.com/time-series-data-visualization-with-python/
3. https://www.kaggle.com/code/subinium/all-you-need-is-time-series-visualization-20
4. https://matplotlib.org/stable/plot_types/index.html
5. https://matplotlib.org/stable/tutorials/images.html#sphx-glr-tutorials-images-py
6. https://dzone.com/refcardz/data-mining-discovering-and#section-1

LAB Experiment -1

Lab Experiment 1

Objectives:

● Familiar with basic supported data structures including list, tuple, dictionary, etc.

● Familiar with basic python packages numpy, pandas and matplotlib packages.

● Data preprocessing.

Task 1: Consider the following given data below and do the followings:

http://archive.ics.uci.edu/dataset/10/automobile

https://archive.ics.uci.edu/ml/machine-learning-databases/autos/ ,!imports-85.data

1. Compute total records in the dataset

2. Compute number of attributes and naming list in the dataset.

3. Display all invalid records in the dataset.

4. Remove all invalid records from the dataset.

5. Compute the average height width and length of all automobiles.

6. Compute the standard deviation height width and length of all automobiles.

7. Describe each attribute in dataset by min, max, average, std and count.

8. Sort all the vehicles by their body-style.

9. Create sample dataset as “sunset-1” by considering the attributes: make, fuel-type, num-of-doors, drive-wheels and engine-location.

10. Create sample dataset as “sunset-2” by considering the attributes: wheel-base, length, width, height and curb-weight.

Lab Experiment 2

Objectives:

- Data preprocessing
- Basic data visualization using Matplotlib package.

Task 1:

Consider the given data below and do the following:

http://archive.ics.uci.edu/dataset/10/automobile

https://archive.ics.uci.edu/ml/machine-learning-databases/autos/ ,!imports-85.data

1. Load the dataset and create the new data frame as MyDF by considering the following attributes: number in () denotes the attribute index number in the data frame.

Symbolling : (1) -3, -2, -1, 0, 1, 2, 3.

fuel-type : (4) diesel, gas.

body-style: (7) hardtop, wagon, sedan, hatchback, convertible.

drive-wheels: (8) 4wd, fwd, rwd.

wheel-base: (10) continuous from 86.6 120.9.

length: (11) continuous from 141.1 to 208.1.

width: (12) continuous from 60.3 to 72.3.

height: (13) continuous from 47.8 to 59.8.

num-of-cylinders: (16) eight, five, four, six, three, twelve, two.

bore: (19) continuous from 2.54 to 3.94.

compression-ratio: (21) continuous from 7 to 23.

horsepower: (22) continuous from 48 to 288.

city-mpg: (24) continuous from 13 to 49.

highway-mpg: (25) continuous from 16 to 54.

price: (26) continuous from 5118 to 45400.

2. Remove all the invalid samples from the dataset

3. Visualize the distribution of all the attributes. Plot Histogram for each attribute.

4. Analyze distribution of all the attributes. Visualize nominal attributes using bar graph and numerical attributes with boxplot.

5. Visualize average height, width and length for each body style.

6. Visualize average height, width and length for each fuel-type.

7. Visualize maximum height, width and length for num-of-cylinders.

8. Visualize the vehicle distributions in terms of their varying prices.

Lab Experiment 3

Objectives:

● Data preprocessing

• Basic data visualization using Matplotlib package.

Task 1: Consider the given data below and do the following:

http://archive.ics.uci.edu/dataset/10/automobile

https://archive.ics.uci.edu/ml/machine-learning-databases/autos/ ,!imports-85.data

1. Load the dataset and create the new data frame as MyDF by considering the following attributes: number in () denotes the attribute index number in the data frame.

• Symbolling : (1) -3, -2, -1, 0, 1, 2, 3.

• fuel-type : (4) diesel, gas.

• body-style: (7) hardtop, wagon, sedan, hatchback, convertible.

• drive-wheels: (8) 4wd, fwd, rwd.

• wheel-base: (10) continuous from 86.6 120.9.

• length: (11) continuous from 141.1 to 208.1.

• width: (12) continuous from 60.3 to 72.3.

• height: (13) continuous from 47.8 to 59.8.

• num-of-cylinders: (16) eight, five, four, six, three, twelve, two.

• bore: (19) continuous from 2.54 to 3.94.

• compression-ratio: (21) continuous from 7 to 23.

• horsepower: (22) continuous from 48 to 288.

• city-mpg: (24) continuous from 13 to 49.

• highway-mpg: (25) continuous from 16 to 54.

• price: (26) continuous from 5118 to 45400.

2. Remove all the invalid samples from the dataset

3. Visualize the vehicles' height, width, and length in a single graph. (line graph)

4. Visualize the vehicles' height, width, and length in a single graph. (plot line graph and all lines have different style and different marker style)

5. Visualize the vehicles' height, width, and length in the increasing order of their price. (line graph)

6. Visualize the relationship between height and width using a scatter plot.

LAB-4

General Instructions

Before you start a lab:

1. Create a notebook file with naming convention “<roll No>_<Lab No>”. 2. Upload the notebook file with a pdf file under Lab Assignment.

Objective:

• Basic data visualization using Matplotlib package.

Exp-1

Consider the data set having 11 employees.

E_age = [25,26,27,28,29,30,31,32,33,34,35]

E_salary = [38496,42000,46752,49320,53200,56000,62316,64928,67317,68748,73752] E_Overhead= [45372,48876,53850,57287,63016,65998,70003,71496,75370,83640,45872]

1. First, describe the uses of a set of graphs below and then plot the graph by considering the given above dataset. (The graph should be complete with all aspects like axis titles, figure titles, legends, etc.)

i) Scatter Plot

ii) Pie Chart

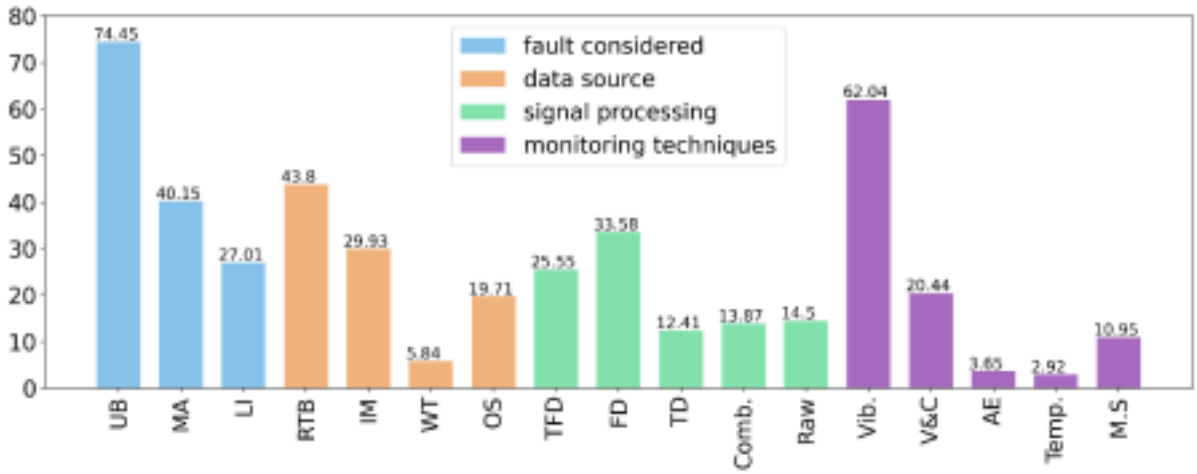
iii) Violin Plot

iv) Heat map

v) Contour Plot

Exp-2

Regenerate this graph using the matplotlib library.



Lab Experiment 5

Objectives:

● Basic data visualization using Matplotlib package.

Task 1: Exp-1

Consider the data set having 11 employees.

E_age = [25,26,27,28,29,30,31,32,33,34,35]

E_salary = [38496,42000,46752,49320,53200,56000,62316,64928,67317,68748,73752]

E_Overhead= [45372,48876,53850,57287,63016,65998,70003,71496,75370,83640,45872]

1. Visualize the data using a bar graph in a single figure.

2. Visualize individual bars for age, salary, and overhead by considering three subplots in a single figure. Arrange all subplots horizontally.

3. Visualize individual bars for age, salary, and overhead by considering three subplots in a single figure. Arrange all subplots vertically.

4. Visualize individual bars for age, salary, and overhead by considering three subplots in a single figure. Arrange all subplots vertically, and they will share the x-axis only.

5. Visualize individual bars for age, salary, and overhead by considering three subplots in a single figure. Arrange all subplots vertically, and they will share the x-axis and y-axis.

6. Visualize the person's age, and salary using a line graph. Consider the y-axis as a logarithmic scale.

7. Visualize the person's age using a bar graph and highlight the person having a minimum age.

8. Consider the data given below and plot in the polar coordinate system.

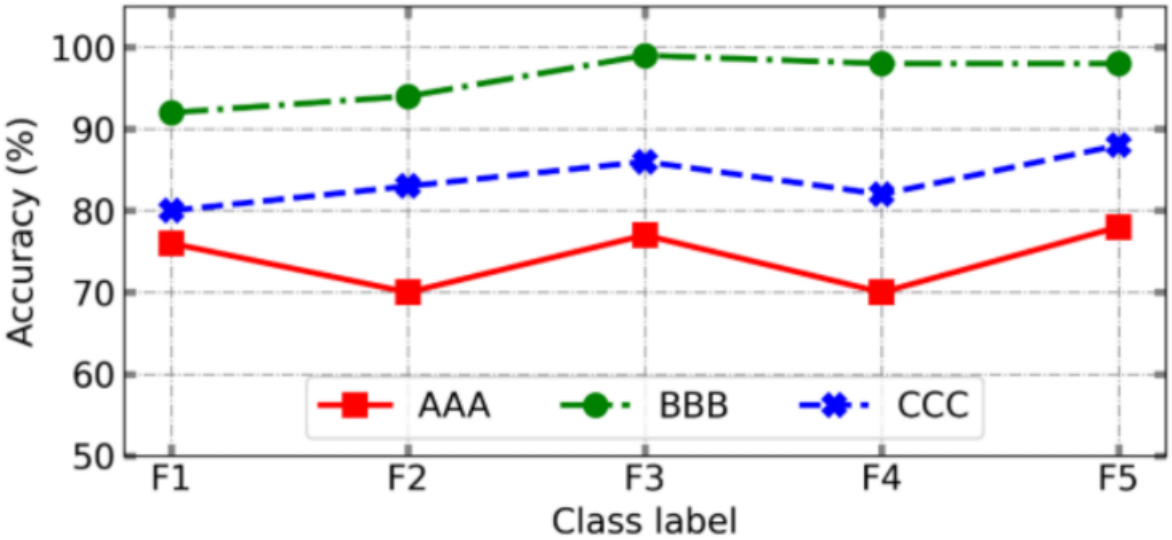
Days = [Mon, Tue, Wed, Thu, Fri, Sat, Sun]

Temp= [12, 14, 18, 20, 17, 13, 11]

Air Quality = [100, 112, 120, 130, 125, 117, 105]

Exp-2

Regenerate this graph using the matplotlib library.



Lab Experiment 6

Objectives:

● Basic data visualization using Matplotlib package.

Task 1: Exp-1

Visualise the data by drawing valid graph

Consider the world population data. (population.csv)

1. Visualize the population of the top 25 countries (population-wise)

2. Visualize the population and middle age of the top 25 countries (population-wise)

3. Visualize the total population, urban population, and Land area of the top 10 countries having the lowest middle age.

4. Visualize the middle age (increasing order) of all the countries using dot plot.

5. Visualize the correlation among following attributes by creating 4 subplots (2x2 Grid):

a. Population and Land area

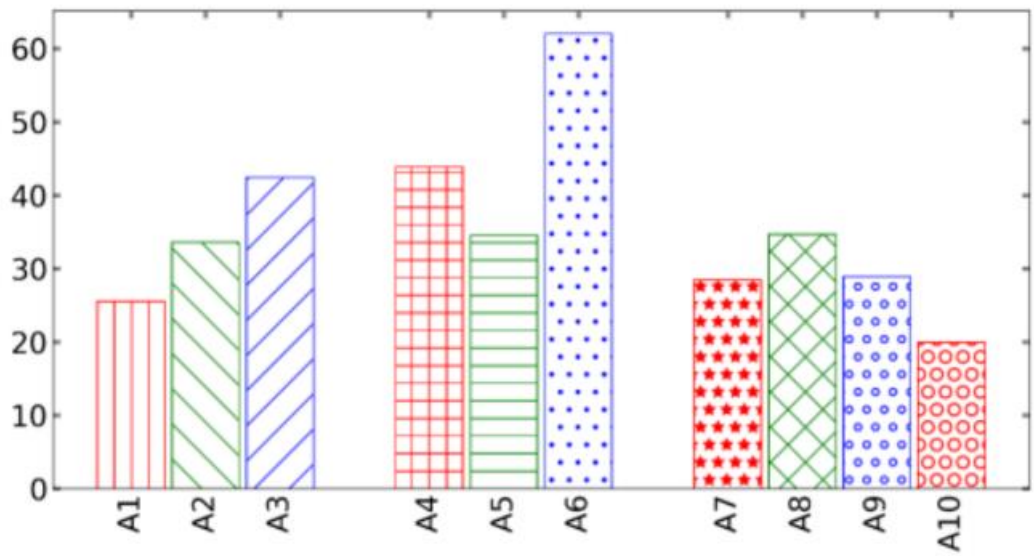
b. Density and land Area

c. Urban Pop and world share

d. Population and Urban Pop

Exp-2

Regenerate this graph using the matplotlib library.

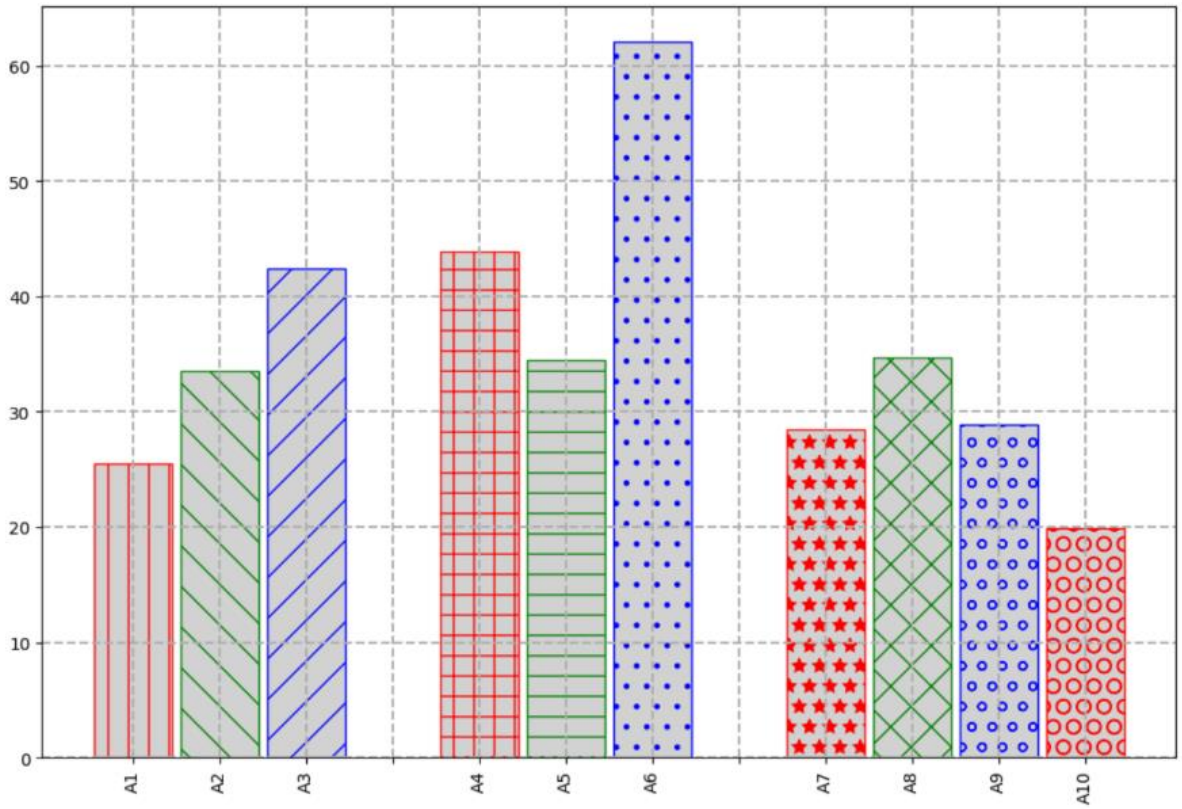


Lab Experiment 7

Objectives:

● Analyse the dataset features through visualization.

Task 1: Generate the following graph.



Task 2: Analyze the dataset through visualization and prepare a report. The report will reflect the concept of the following:

• Figure, figure size, subplots, ticks, grid, markers, legends, axes, scale, save quality image.

Dataset: The classic dataset on the Titanic disaster covers passengers only, not the crew.

Features:

1. survival - Survival (0 = No; 1 = Yes)

2. class - Passenger Class (1 = 1st; 2 = 2nd; 3 = 3rd)

3. name - Name

4. sex - Sex

5. age - Age

6. sibsp - Number of Siblings/Spouses Aboard

7. parch - Number of Parents/Children Aboard

8. ticket - Ticket Number

9. fare - Passenger Fare

10. cabin - Cabin

11. embarked - Port of Embarkation (C = Cherbourg; Q = Queenstown; S = Southampton)

12. boat - Lifeboat (if survived)

13. body - Body number (if did not survive and the body was recovered).

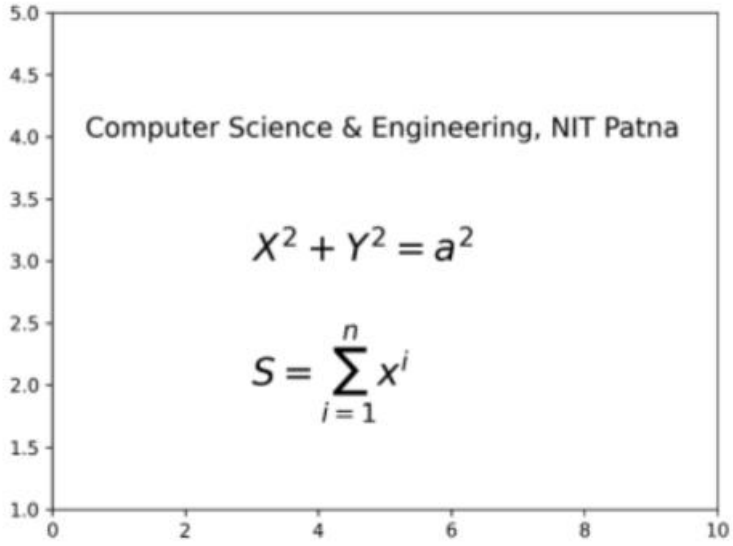
Lab Experiment 8

Objectives:

● Use of color, text, and annotation.

Task 1: Regenerate the following visualizations:

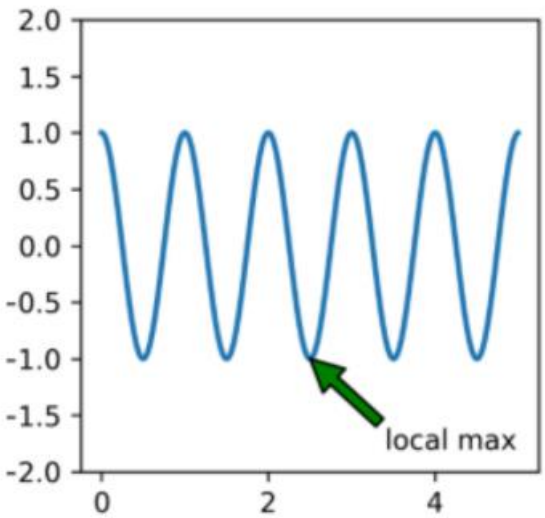
Q.1



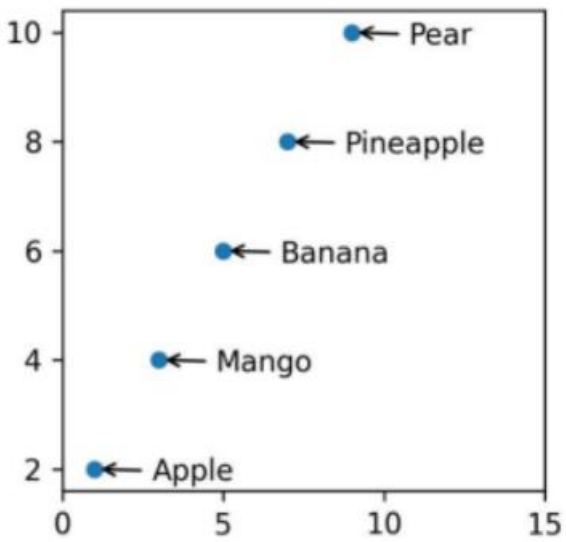
Q2:

t = np.arange(0.0, 5.0, 0.01)

s = np.cos(2*np.pi*t)

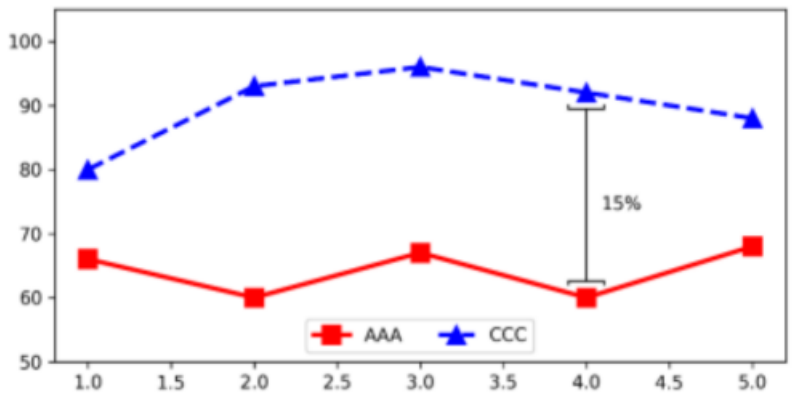


Q3:



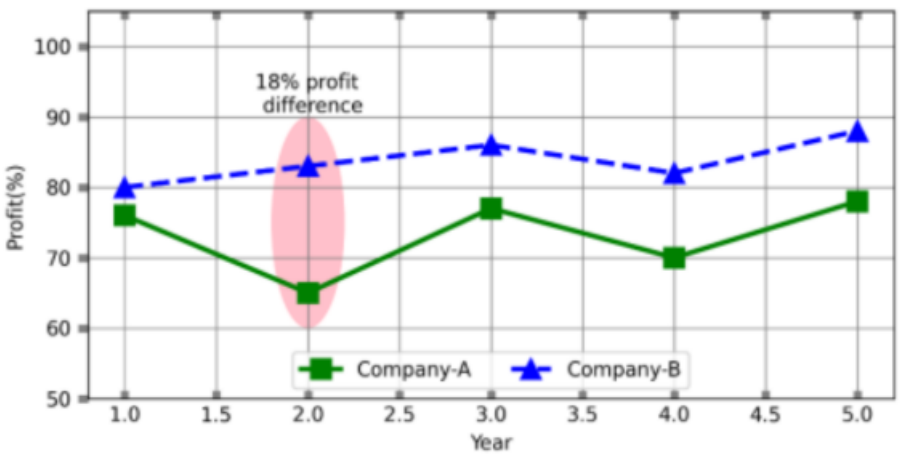
Q4:

data = [[66, 60, 67, 60, 68],[80, 93, 96, 92, 88]]



Q5:

data = [[76, 65, 77, 70, 78],[80, 83, 86, 82, 88]]



Lab Experiment 9

Objectives:

● Create 3-D plots using matplotlib library.

Task 1:

Consider the iris dataset having 150 samples with four features (sepal.length sepal.width petal.length petal.width) and three classes ('Setosa', 'Versicolor', 'Virginica'). Analyze the data through visualization.

Q1. Plot the distribution of all four feature in single 3D plot. (Histogram)

Q2. Plot the distribution of all four feature in single 3D plot. (Density estimation)

Q3. Visualize the distribution with respect to all three classes by considering two features.

Q4. Visualize the distribution with respect to all three classes by considering three features.

Lab Experiment 10

Objectives:

● To understand complex data visualization

● Time series data

● Images data

● Geospatial Data

Task 1:

Q1. Consider the daily minimum temperature dataset and do the following:

Data source: https://raw.githubusercontent.com/jbrownlee/Datasets/master/daily-min-temperatures.csv

• Visualize the day-wise data from 1981 to 1990.

• Visualize the average month-wise data from 1981 to 1990.

• Visualize the average quarter-wise data from 1981 to 1990 and highlight the highest temperature peaks of the quarter with annotation.

• Visualize the minimum daily temperature distribution through violin plot year-wise from 1981 to 1990.

Q2. Do the following to visualize the image:

• Read a colour image and display it in the notebook file.

• Next, create a subplot of grid size 3x3. The first row will display the red, green, and blue channels of an image. The second row will display the histogram of each channel.

Q3. Display the map of India and highlight the region of Uttar Pradesh state.